Double-UV Relative-Momentum Enhancement in the GL00 $\gamma\gamma \to \gamma\gamma$ Integrand

Investigation note

June 14, 2026

Abstract

This note documents the current understanding of a large-weight feature seen in the generated GL00 two-loop integrand for $\gamma\gamma \to \gamma\gamma$ with the nested integrated UV counterterm included. The issue is not an ordinary uncancelled UV divergence in a generic double-UV direction. The problematic samples lie close to a relative-momentum ridge: two sampled loop-momentum-basis (LMB) rays become hard, nearly collinear, and have comparable radii, so that one physical internal momentum is a small difference of the two hard basis momenta. The resulting enhancement is locally integrable, but it creates large Monte-Carlo variance because the current sampling channels do not parameterize that relative momentum directly. LMB multichanneling is not merely a random choice of LMB: it includes a normalized OSE-product prefactor, and that prefactor strongly suppresses inappropriate channels near the edge-small limit. The remaining question is how much this discrete reweighting, and in particular its exponent α, can reduce the variance without an additional continuous relative-momentum map. A fixed-ray option was added to the UV profiler to make this non-generic direction visible in profiling runs.

1 Executive summary

For the regenerated all-LMB setup, GammaLoop reports all eleven loop momentum bases as active channels:

$$(5, 10),\ (4, 10),\ (5, 9),\ (4, 9),\ (5, 8),\ (4, 8),\ (5, 7),\ (4, 7),\ (5, 6),\ (4, 6),\ (4, 5).$$

This required one small display/introspection correction: the display command must read the generated multichannel setup instead of recomputing the old heuristic.

The short all-LMB Monte-Carlo run still found its largest signed evaluations in channels $(5, 10)$ and $(5, 9)$, not in a channel that explicitly uses edge 4 as a basis variable. That is consistent with a variance problem rather than a pure channel-enumeration problem. In the original graph-generation basis,

$$\begin{aligned} e_4 &= K_0, &(1)\\ e_5 &= -K_0 + K_1 + P_0, &(2)\\ e_9 &= K_1, &(3)\\ e_{10} &= -P_2 + K_1 + P_0 + P_1. &(4) \end{aligned}$$

In channel $(5, 10)$, where $u = e_5$ and $v = e_{10}$ are sampled directly,

$$e_4 = v - u + P_2 - P_1. \qquad (5)$$

Thus a sample with u and v both hard, nearly collinear, and of similar radius maps to a much smaller edge-4 momentum. This is precisely the geometric configuration that the OSE channel weights can identify only indirectly: they can favor LMBs containing edge 4, but they do not replace the continuous variables (u, v) by common and relative momenta.

2 What the inspected weight contains

The approach plots below use the value printed by GammaLoop as "The evaluation of integrand". This is the event weight passed to the integrator. The same inspect output also prints the parameterization jacobian separately and a much smaller "integrand result" before multiplication by that jacobian. Therefore the plotted quantity already includes the spherical parameterization jacobians and the conformal-map jacobians. The large-radius slope in the plots is a statement about the actual weight seen by the integrator, not merely about the unweighted analytic integrand.

3 Minimal toy model

The minimal structure needed to reproduce the observed mechanism is not a full two-loop numerator. It is enough to keep:

- two hard sampled LMB momenta u and v;
- a physical internal momentum $q = v - u + Q$ that is a difference of those hard momenta plus a fixed external shift Q;
- the post-subtraction and parameterization effect that leaves a positive power of the common hard radius in the event weight;
- an energy denominator sensitive to the small relative momentum.

The toy event weight used for the figures is

$$W_{\text{toy}}(u, v) = C(\hat{u}, \hat{v}, \rho)\,\frac{R^2}{E_q}, \qquad E_q = \sqrt{|v - u + Q|^2 + q_0^2}, \qquad R = \frac{|u| + |v|}{2}, \tag{6}$$

where $\rho = |v| \,/\, |u|$ and C is a smooth bounded angular factor. The constant q_0 regulates the exactly coincident toy edge and stands in for the fact that the full GL00 expression has masses, external shifts, and numerator cancellations. The power R^2 should not be read as a universal UV degree. It is the minimal effective power required to match the observed weighted large-radius approach after all jacobians are included.

Let the relative separation scale as

$$|v - u + Q| \simeq \epsilon R \tag{7}$$

for a fixed but small opening/radius mismatch ϵ. Then

$$W_{\text{toy}} \simeq \frac{R^2}{\epsilon R} = \frac{R}{\epsilon}. \tag{8}$$

This gives a slope-one growth along a ray where the two UV momenta are scaled together while their relative mismatch is held fixed. If the two hard momenta coalesce more quickly than $1/R$, the toy model crosses over toward R^2/q_0. For generic separated directions the coefficient is much smaller; the bad region is the ridge where $\rho \simeq 1$ and $\theta(u, v) \simeq 0$.

The singularity is locally integrable in the relative momentum. In relative variables $\mathrm{d}^3 q = |q|^2 \,\mathrm{d}\,|q|\,\mathrm{d}\Omega_q$, a $1/\,|q|$ factor gives

$$\mathrm{d}^3 q\,\frac{1}{|q|} \sim |q|\;\mathrm{d}\,|q|\;\mathrm{d}\Omega_q,$$

so the local integral is finite. Monte Carlo variance can nevertheless be large, especially when the sampling variables are the hard momenta u and v rather than the common momentum and the relative momentum q.

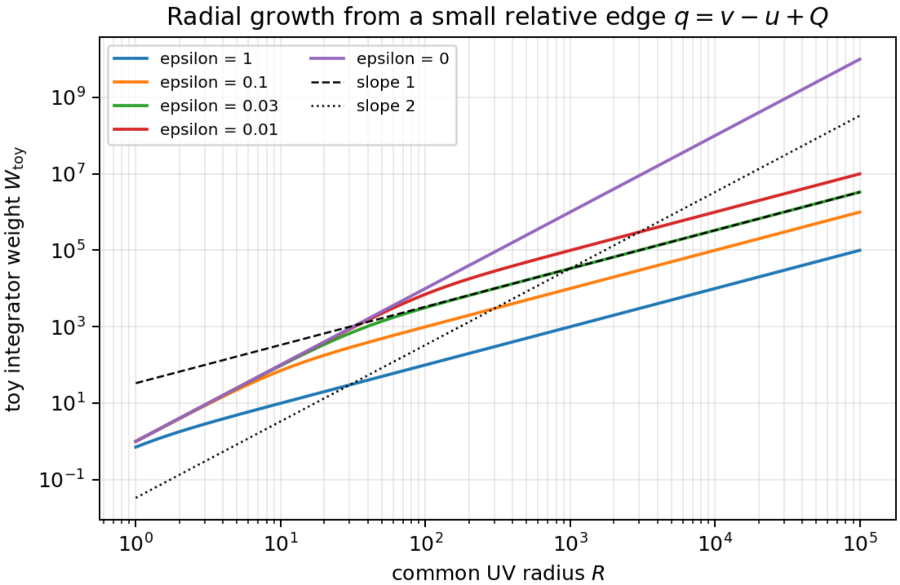

Figure 1: Toy-model radial scaling. For fixed nonzero relative mismatch ϵ, the weighted integrand approaches a slope-one growth, $W_{\text{toy}} \sim R/\epsilon$. Exact coalescence crosses over to slope two in this deliberately simplified model.

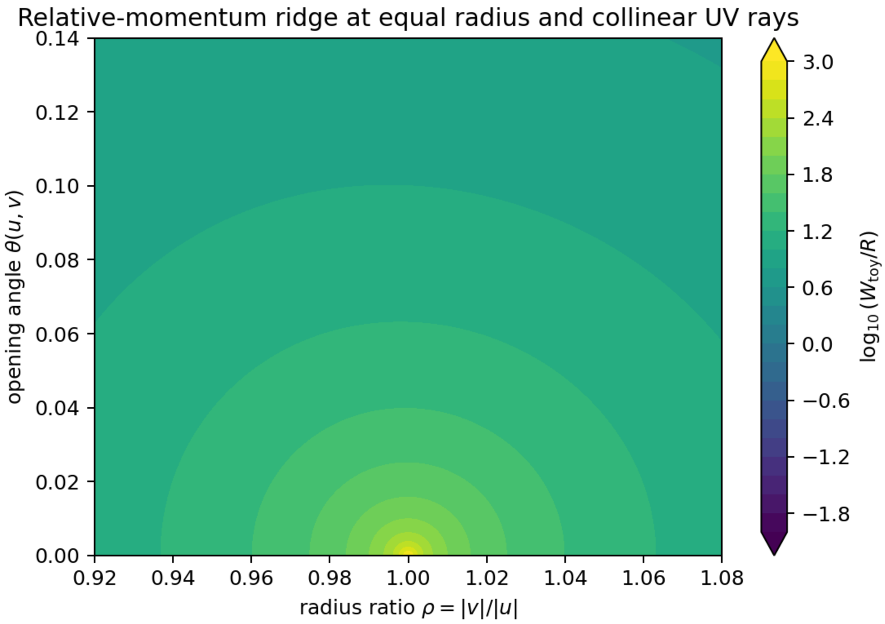

Figure 2: Toy relative-momentum ridge at fixed large common radius. The enhancement is concentrated near equal radius, $\rho = 1$, and small opening angle. Existing LMB channels can reduce coefficients, but they do not create a direct sampling variable for this ridge.

4 GL00 approach data

The GL00 radial scan starts from the high-weight point

$$x_A = (0.9938195811740000,\ 0.6303276140313304,\ 0.6037887485521857,\\ 0.9939524818481166,\ 0.6277285639451646,\ 0.6151209153931045).$$

At $\lambda = 1$, the two radial variables correspond to approximately

$$r_0 \simeq 4.8240 \times 10^4, \qquad r_1 \simeq 4.9307 \times 10^4.$$

The two radii are therefore both in the UV and within about two percent of each other. The angular coordinates are also close; the fitted opening angle in the angular scan is

$$\theta_A \simeq 2.82 \times 10^{-2}.$$

The large-radius fit of the GammaLoop inspected weight gives:

case	fit range	slope in $\log \lvert W\rvert$ vs. $\log \lambda$	points
LMB multichannel off	$\lambda \geq 1$	1.00005	27
LMB multichannel on	$\lambda \geq 1$	1.00424	27

The coefficient changes strongly when LMB multichanneling is enabled, but the large-radius power does not. This is the important observation: the current multichanneling can reduce the normalization of the problematic ray, but it does not alter the radial power seen by the integrator.

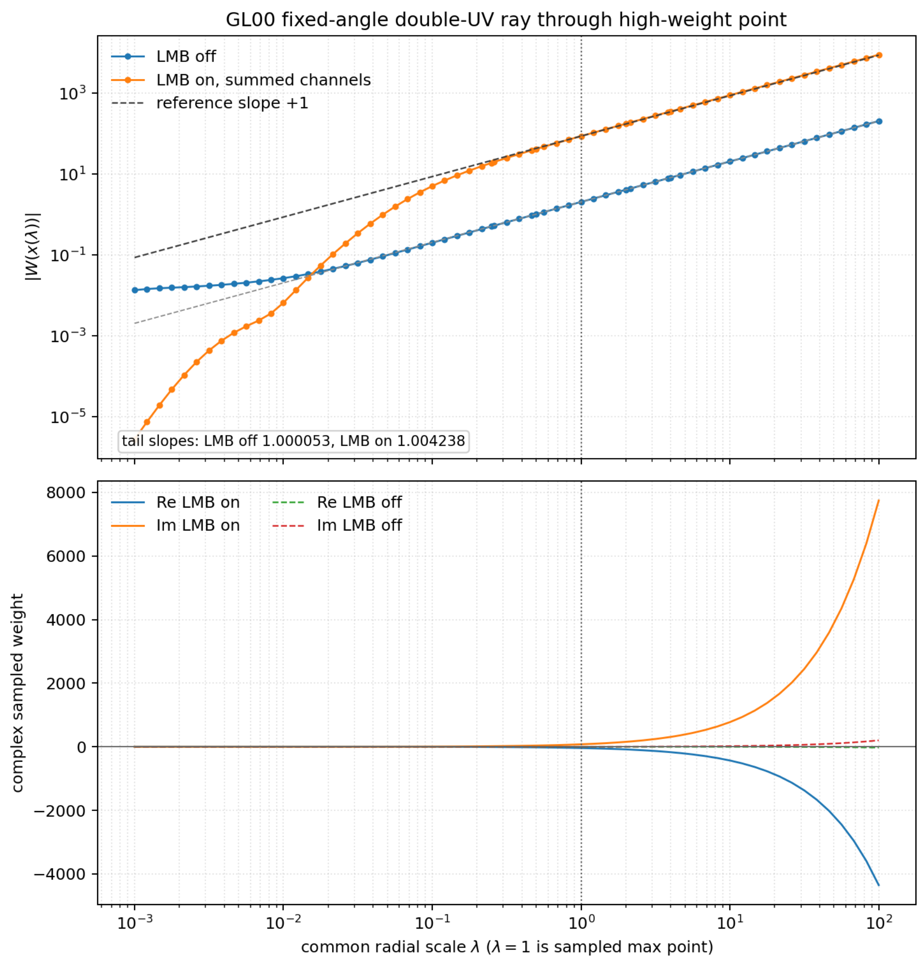

Figure 3: GammaLoop inspected event weight along the high-weight double-UV ray. All parameterization jacobians are included. The large-radius behavior is approximately linear in the common UV scale.

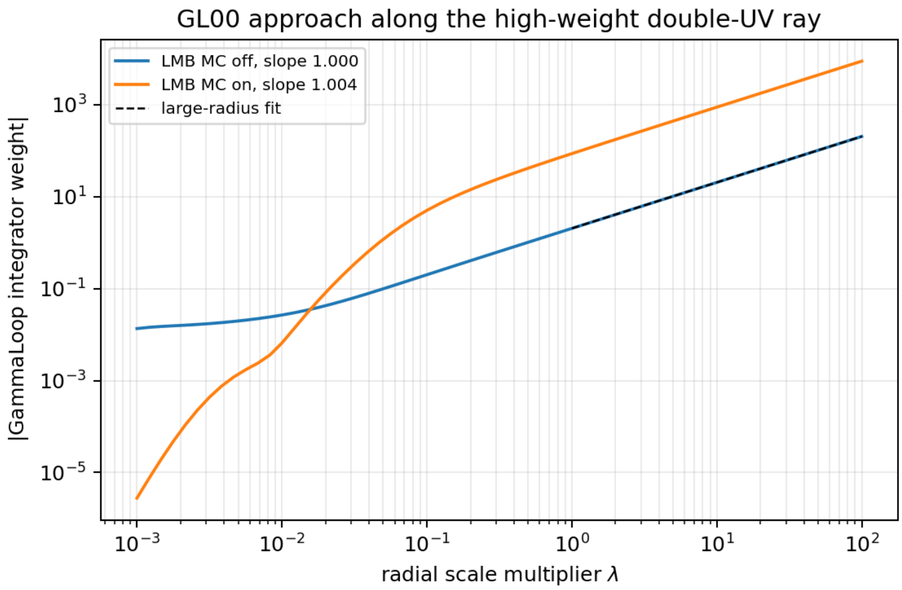

Figure 4: Compact replot of the same radial data, with large-radius fit slopes. The fit is not intended to model the low-radius region.

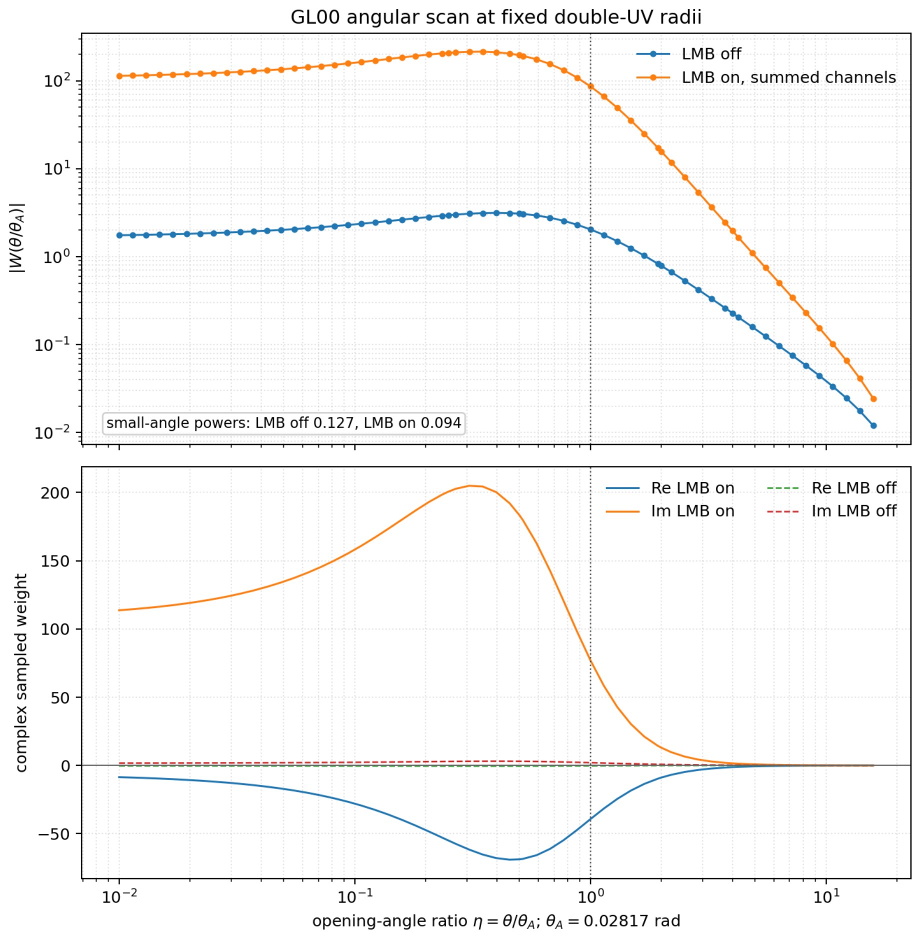

Figure 5: Angular approach scan around the same GL00 point. The coefficient is highly sensitive to the relative direction of the two hard rays, supporting the interpretation as a relative-momentum ridge rather than a generic double-UV failure.

5 Why the standard UV profile misses the enhancement

The ordinary `profile ultra-violet` command samples one random three-momentum per LMB loop variable and then rescales the selected subset. It therefore measures the generic large-momentum degree of divergence for that sampled ray. The relative-momentum ridge discussed above is a correlated configuration: two UV rays must be nearly parallel and have nearly equal radii, so that a physical edge momentum such as $v - u + Q$ stays anomalously small compared to the two hard sampled basis momenta. A random UV profile ray hits this condition with probability zero.

There is also a conceptual difference between the UV profiler and the integration-weight plots. The profiler works directly in momentum space and multiplies by the UV scaling measure factor used for the degree-of-divergence test. It does not include the full x-space parameterization jacobian, adaptive discrete-channel probability, or the conformal-map density that define the event weight seen by the integrator. This is why the standard UV profile can be clean while the integrator still sees a high-variance, locally integrable ridge.

To make the correlated direction testable, the profiler now accepts fixed UV ray data:

```
profile ultra-violet ... \
  --uv-ray-directions dx0 dy0 dz0 [dx1 dy1 dz1 ...] \
  --uv-ray-norms n0 [n1 ...]
```

One direction or norm is repeated for all loop variables; otherwise one three-vector and one norm can be supplied per loop variable. If directions are given but no norm is supplied, the default starting norm is E_{cm}.

The same profiler was then run on the fixed ray matching the angular direction and radius ratio of the GL00 approach scan:

```
--uv-ray-directions
  -0.66816740923660423 -0.71446756114601095 0.20757749710437134
  -0.67621171790860379 -0.69980455265688968 0.23024183078620908
--uv-ray-norms 1 1.0221114989619468
```

The summed, not-per-orientation UV profiles are:

profile	entries	worst slope	r^2	bad entries	arb retry
random ray, all subsets	33	-0.997272	0.999997	0/33	0/33
tailored ray, double-free subsets	11	-0.812993	0.986953	11/11	11/11
tailored ray, single-free subsets	22	-1.000429	1.000000	0/22	0/22

Here "bad" means a fitted slope above the profiler warning threshold -0.9. The random-ray result is the expected generic UV profile. With the tailored ray, all double-free subsets move to the same fitted behavior,

$$\log |I_{\mathrm{prof}}| \simeq -0.812993 \log s + \mathrm{const}, \qquad r^2 = 0.986953,$$

and all of those double-free fits require the arbitrary-precision retry. Single-free subsets remain ordinary.

A follow-up scan continuously shrinks the plot-ray mismatch. Let $a = 1$ denote the GL00 plot ray above and $a = 0$ denote the exactly collinear, equal-radius ray with the first plot-ray direction used for both loop variables. The second direction is

$$\hat{v}(a) = \frac{(1-a)\hat{u} + a\hat{v}_{\mathrm{plot}}}{|(1-a)\hat{u} + a\hat{v}_{\mathrm{plot}}|}, \qquad \rho(a) = 1 + a(\rho_{\mathrm{plot}} - 1).$$

The table reports the summed double-free UV-profile slope. All 11 double-free LMB subsets give the same number for each row.

a	$\theta(\hat{u}, \hat{v})$	$\rho - 1$	$10^3 \to 10^5$	$10^3 \to 10^7$
1.000	2.817×10^{-2}	2.211×10^{-2}	-0.814	-0.931
0.500	1.408×10^{-2}	1.106×10^{-2}	-0.684	-0.881
0.200	5.633×10^{-3}	4.422×10^{-3}	-0.434	-0.776
0.100	2.817×10^{-3}	2.211×10^{-3}	-0.182	-0.661
0.050	1.408×10^{-3}	1.106×10^{-3}	$+0.104$	-0.516
0.020	5.633×10^{-4}	4.422×10^{-4}	$+0.471$	-0.289
0.010	2.816×10^{-4}	2.211×10^{-4}	$+0.699$	-0.101
0.005	1.408×10^{-4}	1.106×10^{-4}	$+0.860$	$+0.092$
0.000	0	0	$+1.000$	$+1.000$

A sparse extension to 10^9 gives slopes -0.964 for $a = 1$, -0.820 for $a = 0.1$, -0.611 for $a = 0.02$, -0.376 for $a = 0.005$, and $+1.000$ for $a = 0$. This is the expected non-uniform limit. For any fixed nonzero angular/radius mismatch, probing deep enough eventually resolves the relative momentum and pushes the fit back toward the generic negative UV behavior. As the mismatch is reduced, the crossover moves to larger scales, and over the finite UV windows relevant to the integrator the fitted slope degrades through zero and becomes positive. The exactly coincident GL00-direction ray remains positive because the relative hard momentum is kept fixed to the external shift instead of growing with the common UV scale.

The same two profiler runs were also repeated with `--per-orientation`. For the random ray, the double-free per-orientation profile remained clean: the worst double-free orientation slope was -0.988527, with no arbitrary-precision retries and no slope above -0.9. For the tailored ray, the issue is not isolated to a unique orientation. Among the 62 fitted orientations, 48 have at least one double-free entry above -0.9. The strongest orientations are:

orientation	max slope	avg. double-free slope	bad entries	arb retry
`0000+++--++`	-0.614883	-0.614883	11/11	11/11
`0000--+--+-`	-0.615094	-0.615094	11/11	11/11
`0000+++--+-`	-0.632973	-0.632973	11/11	11/11
`0000--+--++`	-0.633049	-0.633049	11/11	11/11
`0000---++--`	-0.709478	-0.709478	11/11	11/11
`0000++-++-+`	-0.709620	-0.709620	11/11	11/11

The repeated value across all 11 double-free LMB subsets is expected for this fixed ray because the ray is supplied in each LMB basis. Orientation `0000+++--++` is the worst fitted one, but it is only slightly worse than `0000--+--+-`; the enhancement is therefore a broad orientation-sector feature rather than a single-orientation defect.

The z-axis exactly collinear equal-radius test ray, $(0,0,1)$ and $(0,0,1)$ with norms $1,1$, produced missing fits rather than stable slopes. This was a direction-specific numerical special case, not the generic behavior of coincident rays; the GL00 plot-direction coincident ray gives the stable positive slopes shown above.

These fixed-ray results are not evidence for a generic uncancelled UV divergence. They instead expose a non-uniform, codimension relative-momentum ridge that the random-ray UV profile hides.

This is not in contradiction with the positive-slope event-weight plots. The UV profile uses momentum-space samples with `integrator_weight=1` and multiplies the raw integrand by only the scale-measure factor $s^{3n_{\mathrm{UV}}}$. For the double-UV tests here this is s^6. The radial approach plots, by contrast, show the density seen by the integrator after the spherical linear map. For one hard loop,

$$r = E_{\mathrm{cm}} b \, \frac{x}{1-x}, \qquad r^2 \frac{\mathrm{d}r}{\mathrm{d}x} \sim r^4,$$

whereas the UV shell-measure factor used by the profiler is r^3 per loop. Thus the plotted x-space weight contains one extra large-radius power per hard loop compared to the profiler's DOD diagnostic. A tailored-ray profiler slope near -0.813 can therefore correspond to a pointwise x-space weight that grows roughly like a positive power along the same radial approach. When the rays are tuned even closer together, the profiler itself finds zero or positive slopes along that forced line. The transverse angular and radius-ratio volume is not represented by this one-dimensional profile, so the positive forced-ray DOD identifies an integrable relative singularity and a variance problem rather than a generic UV divergence of the full integral.

The profiler JSON and logs for the default, plot-ray, simplified split-ray, exactly collinear, and per-orientation tests are saved under `DoubleUVIntegrableSingularity/data/uv_profile_*`.

6 Current long-run integration checks

The current all-LMB one-million-sample diagnostic, using Monte Carlo over LMB channels, kinematics A, helicity $+-+-$, $m_{\mathrm{UV}} = 9.188$, and $\mu_R = 91.188$, gave:

component	value	error	relative error
real	-6.8192×10^{-6}	4.7659×10^{-6}	69.9%
imag	-5.4228×10^{-6}	3.3372×10^{-6}	61.5%

The largest signed weights in that run were:

component/sign	max evaluation	LMB channel
real +	$+5.3591 \times 10^{-1}$	$(5,9)$
real −	-4.3118	$(5,10)$
imag +	$+5.4193 \times 10^{-1}$	$(5,9)$
imag −	-2.5294	$(5,10)$

A longer Monte-Carlo-over-LMB run with 90 M samples and 20 cores completed in 739.3 s. It did not reach the requested sub-10% Monte-Carlo error:

component	value	error	relative error	max-weight impact
real	-4.4150×10^{-7}	4.2964×10^{-7}	97.3%	0.791
imag	-1.2004×10^{-6}	3.0318×10^{-7}	25.3%	0.134

The largest signed weights in this run moved to LMB channel $(5,7)$. The largest real negative event was -31.4201, and the largest imaginary negative event was -14.4356, both at the same sampled point. The adaptive grid probability for channel $(5,7)$ was about 42.2%, but the continuous relative ridge still dominated the variance.

An explicit summed-LMB run with 10 M samples was run for a comparable wall time, 740.9 s. Summing removes the discrete LMB sampling noise but makes each sample roughly an order of magnitude more expensive. At equal wall time it also did not converge:

component	value	error	relative error	max-weight impact
real	$+6.6162 \times 10^{-8}$	3.2102×10^{-7}	485.2%	1.513
imag	-1.1342×10^{-6}	4.4162×10^{-7}	38.9%	0.190

The summed-channel run lowers the absolute largest sampled weights compared to the MC-over-LMB run, but the real part is close enough to zero that its relative error remains very large. These results show that the default $\alpha = 3$ LMB multichannel setup did not reach stable sub-10% errors in the tested wall time. They should not be read as proof that LMB multichanneling is irrelevant, or that no choice of α can help; the unresolved issue is the continuous relative-momentum ridge left after the discrete OSE reweighting.

The saved run artifacts are in the report data directory, with separate JSON and log files for the 90 M MC-over-LMB run and the 10 M summed-LMB run.

7 What the LMB multichannel prefactor does

The generated all-LMB setup contains channels where edge 4 is a basis edge, and the LMB multichanneling code does use this information. For a channel c with basis edges $e \in c$, the channel weight is

$$w_c = \left(\prod_{e\in c} E_e\right)^{-\alpha}, \qquad p_c = \frac{w_c}{\sum_j w_j}. \tag{9}$$

With `lmb_channels = "monte_carlo"`, one LMB is sampled with this normalized weight. With `lmb_channels = "summed"`, all generated LMBs are evaluated with the same normalized prefactors. The default value is $\alpha = 3$, and the flat `[sampling]` parser now exposes this as an `alpha` field so that it can be scanned directly.

I evaluated this normalized prefactor on the same ray family used in the fixed-ray UV-profile scan. The table reports the raw probability assigned to the problematic $(5,10)$ interpretation and the total probability assigned to channels containing edge 4:

a	R	α	E_4/R	$p_{(5,10)}$	$\sum_{4\in c} p_c$
1.000	10^3	1	2.041×10^{-1}	3.050×10^{-2}	0.854066
1.000	10^3	3	2.041×10^{-1}	1.621×10^{-3}	0.992885
1.000	10^7	1	3.566×10^{-2}	5.814×10^{-3}	0.970933
1.000	10^7	3	3.566×10^{-2}	7.726×10^{-6}	0.999961
0.100	10^7	1	3.584×10^{-3}	5.959×10^{-4}	0.997020
0.100	10^7	3	3.584×10^{-3}	7.687×10^{-9}	1.000000
0.020	10^7	1	7.206×10^{-4}	1.200×10^{-4}	0.999400
0.020	10^7	3	7.206×10^{-4}	6.240×10^{-11}	1.000000
0.005	10^7	1	1.843×10^{-4}	3.072×10^{-5}	0.999846
0.005	10^7	3	1.843×10^{-4}	1.044×10^{-12}	1.000000
0.000	10^7	1	1.973×10^{-5}	3.287×10^{-6}	0.999984
0.000	10^7	3	1.973×10^{-5}	1.279×10^{-15}	1.000000

The conclusion is that the OSE prefactor is doing something very strong. On the near-collinear rays, default $\alpha = 3$ assigns essentially all normalized channel weight to LMBs containing edge 4 and suppresses the direct $(5, 10)$ interpretation by many orders of magnitude. Increasing α would make this channel selection even sharper; decreasing it would make the channel selection smoother. Therefore the previous statement that LMB MC merely samples an LMB, or that it cannot be sufficient by construction, would be too strong.

What remains true is that this prefactor is a discrete LMB reweighting, not a continuous change of variables to

$$P_{\text{common}} \simeq \frac{u+v}{2}, \qquad q = v - u + Q.$$

The problematic region is thin in the original hard-momentum variables and natural in (P_{common}, q) variables. Once the mismatch is fixed and all relevant OSEs scale linearly with the common hard radius, changing α changes the coefficient and channel allocation but not the asymptotic radial power along that ray. Near the exact edge-small limit it can be decisive, because E_4/R becomes tiny and the normalized weight overwhelmingly selects edge-4 channels.

This explains why the longer all-LMB Monte-Carlo run assigned large probability to channel $(5, 7)$, about 42.2%, and sizable probabilities to channels such as $(5, 10)$, about 15.8%, while individual edge-4 channels were only about 1.4% each. The adaptive procedure sees the contribution and variance of the existing channels, but it does not infer a new relative-momentum map. The current data are therefore compatible with two possibilities: default $\alpha = 3$ may simply need more statistics or tuning, or the variance may require an explicit relative-momentum channel in addition to the OSE-prefactor LMB selection. An α scan should be part of the next integration campaign now that the setting is exposed in `[sampling]`.

8 Implications

The current evidence points to a high-variance, locally integrable relative-momentum singularity rather than a wrong integrated UV counterterm or a missing local UV cancellation. The practical options suggested by the toy model are:

1. add a specialized channel or remapping for hard-hard relative momentum ridges, using variables close to (P_{common}, q);

2. scan the LMB prefactor exponent α; larger values focus more strongly on channels containing the small OSE, while smaller values avoid over-focusing and may improve other regions;

3. use explicit summed LMB channels as a diagnostic, while recognizing that this reduces discrete-channel variance but does not by itself remap the continuous relative-ridge variables.